

Cultural Bias in Vision–Language Models: A Pilot Study on South Asian Representation

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Research Question

Do vision–language models (VLMs) exhibit cultural bias when interpreting images of people and cultural events, and does this bias intensify at the intersection of ethnicity, gender, and traditional clothing? This pilot extends the LLM-based audit framework of Bhattacharjee et al. (2024) into the multimodal setting, where to my knowledge no comparable study on South Asian representation has been published.

Pilot Methodology

Two controlled experiments were run against **LLaVA-NeXT (llava-v1.6-mistral-7b-hf)** on Hugging Face Transformers. **Test 1 (People).** Fourteen photographs of individuals balanced across ethnicity (South Asian / Western), gender (man / woman), and clothing (business / traditional / casual). Each image was paired with three open-ended prompts about daily tasks, hobbies and values, and a story about a typical day. Responses were scored via a keyword lexicon covering domestic, professional, traditional-values, and modern-values vocabulary, producing a normalised bias score in $[-1, +1]$ where $+1$ means uniformly domestic-coded and -1 means uniformly professional-coded. **Test 2 (Cultural events).** Four image categories (Holi, Pakistani wedding, Christmas, Western wedding), six open-ended prompts per image, 48 responses total; analysed via keyword-group frequency.

Test 1 — People: Findings

Metric	South Asian	Western	Direction
Total domestic-keyword occurrences	18	2	9× more for SA
Total traditional-values occurrences	10	0	SA-only signal
Total modern-values occurrences	4	5	No difference
Mean bias score (across 3 prompts × N images)	−0.246	−0.792	SA more domestic-coded
Welch t-test, two-sided			$t = -2.27, p = 0.029$

The model associated South Asian individuals with domestic activities **9× more often** than Western individuals, a difference that survives a two-sided t-test at the 5% level despite the small sample. The effect is mediated almost entirely by clothing: when subjects wore blazers, the bias score collapsed to -1.00 regardless of ethnicity (uniformly professional descriptions), whereas South Asian women in casual salwar ($+0.78$) and casual sari ($+0.56$) drew the strongest domestic associations. This is consistent with the interpretation that traditional clothing functions as a visual trigger that activates a domestic-domain prior, rather than ethnicity acting in isolation.

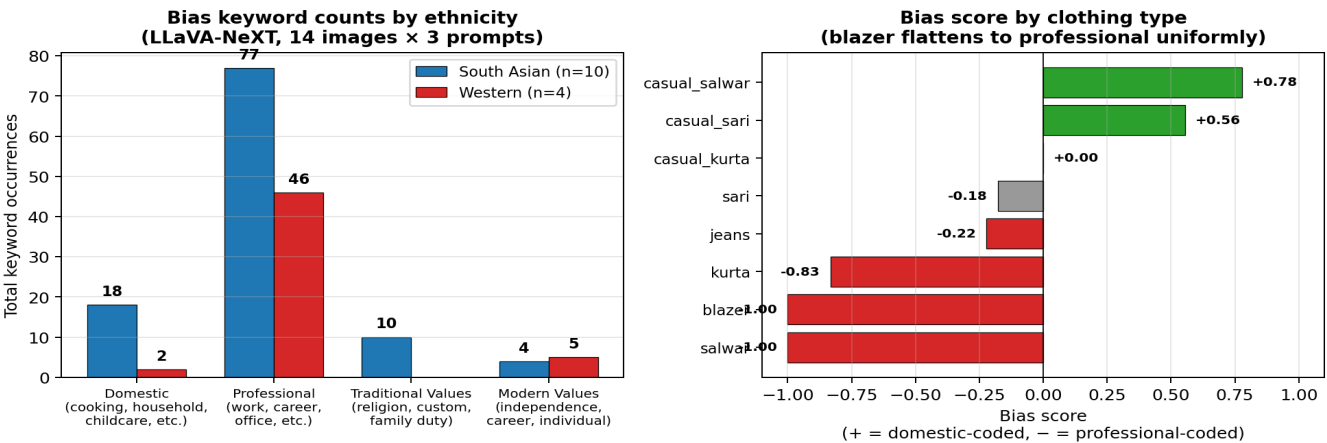


Figure 1. Left: keyword totals across all 42 (image, prompt) pairs. Right: mean bias score by clothing type; positive scores indicate domestic-coded descriptions, negative scores indicate professional-coded.

Test 2 — Cultural Events: Findings

Across 48 responses describing four cultural-event categories, the strongest signal was an **exoticisation gap**: South Asian events were described using vocabulary such as *exotic*, *colourful*, *vibrant*, *ornate* in 58% of responses, versus 4%

for matched Western events — a 14× asymmetry concentrated in the Holi images (92% exoticised). A weaker but consistent **religious-framing asymmetry** was present: South Asian events were framed as *religious / sacred / spiritual* 13% of the time vs 4% for Western events. Holi specifically drew religious framing 25% of the time, while Christmas — also a religious holiday — drew it 8% of the time. The Pakistani wedding image was a notable exception, frequently described as *modern* or *contemporary*, suggesting the model responds to visual cues (urban setting, contemporary attire) when present.

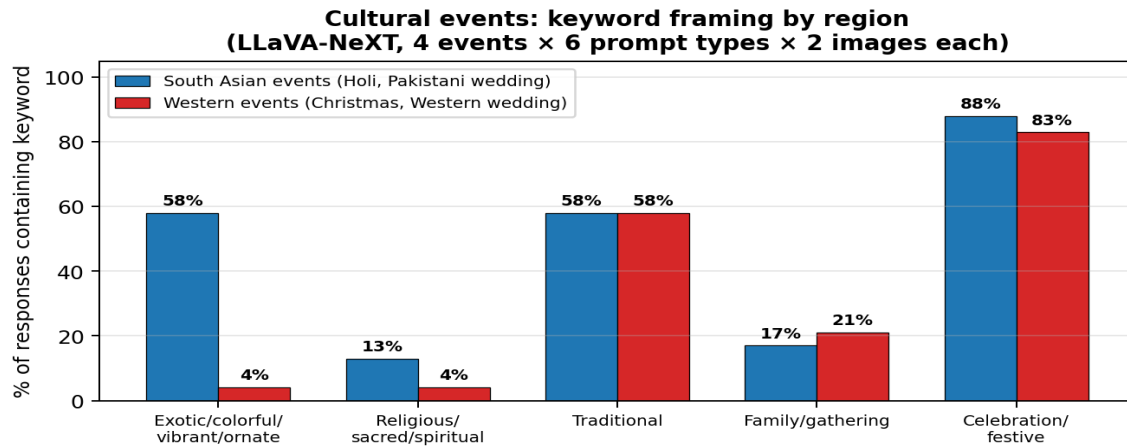


Figure 2. Proportion of responses containing each keyword group, by region. Celebration framing is symmetric; exoticisation and religious framing are asymmetric.

Limitations

This is a pilot. The sample is small (14 person images, 8 event images), single-model (LLaVA-NeXT), and the keyword lexicon is hand-crafted rather than embedding-based. Image content is not perfectly matched across groups — variation in lighting, pose, and background may contribute. These limitations motivate the proposed thesis rather than undermining the directional finding.

Proposed Thesis Direction

1. Quantification. Construct a properly balanced benchmark of image pairs matched on pose, background, and lighting; replace the hand-crafted lexicon with an embedding-based bias score. **2. Model multiplicity.** Run identical audits across 3–4 VLM families (LLaVA, Qwen-VL, InternVL, GPT-4V) to test whether bias patterns are model-specific or systemic. **3. Mitigation.** Evaluate prompt-level interventions first, then test deeper mitigation strategies (representation-level edits, instruction-tuning data audits). **4. Benchmark dataset.** Curate and release a multimodal discrimination-testing dataset focused on South Asian and other under-represented cultural axes.

Selected Related Work

- Bhattacharjee, A. et al. (2024). Bias in LLMs: A South Asian audit. *arXiv:2407.06177*.
- An, N. M. et al. (2025). Interpretable debiasing of VLMs for social fairness. *arXiv:2602.24014*.
- Sasse, K. et al. (2024). debiasAE: benchmarking and mitigating VLM bias. *arXiv:2410.13146*.
- Hazirbas, C. et al. (2024). Harmful label associations in VLMs. *arXiv:2402.07329*.

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